Prediction of PCR and RFS using Machine Learning for Breast cancer

Abstract— **Breast cancer stands as the foremost cancer affecting women in the United Kingdom. To diminish advanced tumors, chemotherapy is frequently employed before surgical intervention. Nevertheless, this treatment harbors toxic effects and does not guarantee universal efficacy. The complete eradication of the tumor at the surgical stage, termed pathological complete response (PCR), significantly enhances the prospects of a cure and extends the duration of relapse-free survival (RFS). RFS denotes the period post-primary cancer treatment during which the patient remains free from any cancer manifestations. Regrettably, a mere 25% of those treated with chemotherapy attain PCR, leaving a majority with persistent disease and varying prognoses. Enhanced prediction of PCR and RFS prior to initiating chemotherapy could lead to more precise patient categorization and improved treatment approaches.**

The methods employed consist of Convolutional neural networks (CNN), Decision trees, Support vector machines (SVM), Logistic regression to achieve a best overall classification accuracy using for PCR (Classification) and a best mean absolute error for RFS (Regression).

Index Terms—**Machine Learning, Classification, Regression, Supervised and Unsupervised Algorithms**

I. Introduction

Breast cancer remains a formidable challenge in healthcare, demanding innovative approaches to enhance treatment efficacy and patient outcomes. In 2020, breast cancer led to the death of 685,000 individuals worldwide. About half of all instances of breast cancer occur in women without particular risk factors, except for their gender and age [1]. Every year in the UK, approximately 55,900 new cases of breast cancer are reported, equating to over 150 cases each day. [2].

The age-standardized mortality rate for breast cancer in high-income countries decreased by 40% from the 1980s to 2020. Countries that effectively lowered breast cancer mortality achieved an annual reduction rate ranging from 2% to 4% per year [3].

Included in the analysis were 31 studies that satisfied the inclusion criteria, with the majority being published after 2013. The predominant machine learning methods employed were decision trees (found in 19 studies, or 61.3%), artificial neural networks (featured in 18 studies, or 58.1%), support vector machines (utilized in 16 studies, or 51.6%), and ensemble learning (employed in 10 studies, or 32.3%). The median sample size encompassed 37,256 patients (ranging from 200 to 659,820), and the median number of predictors was 16 (ranging from 3 to 625). Accuracy across 29 studies varied from 0.510 to 0.971, while sensitivity in 25 studies ranged from 0.037 to 1. Specificity in 24 studies ranged from 0.008 to 0.993, and the area under the curve (AUC) in 20 studies spanned from 0.500 to 0.972. Precision in 6 studies ranged from 0.549 to 1 [3]. All models underwent internal validation, with only one undergoing external validation. validated. Machine learning techniques can significantly improve the accuracy of predicting Pathological Complete Response (PCR) and Recurrence-Free Survival (RFS). This report examines the application of machine learning methods in forecasting PCR and RFS among breast cancer patients. The investigation encompasses the utilization of both clinically assessed attributes and characteristics derived from MRI scans conducted before initiating chemotherapy. Additionally, the report assesses the efficacy of various machine learning methods and offers suggestions for future research endeavors.

II. Methods

Dataset

The dataset consists of a list of 400 patients with their unique id. In this dataset, every patient has 10 clinical characteristics like age, ER, PgG, and HER2, Grade of Chemotherapy, Triple Negative Status, and Tumor Histology Type, Lymph Node Status, Proliferation, and Tumor Stage and 117 features derived from images.

Feature Selection

Feature selection is crucial to the algorithm's successful training. Dedicated feature selection was used like Lasso which reduces the coefficients of minor features, Lasso also lessens the model's sensitivity to noise and anomalies in the training set. In addition to enhancing generalization performance on unobserved data, this helps avoid overfitting.

Recursive feature selection successfully lowers the feature count, avoiding overfitting and enhancing the generalizability of the model. It aids in avoiding the phenomenon known as the "curse of dimensionality," which occurs when a model's complexity and instability rise with the number of features.

Forward feature selection, large datasets with lots of features can be handled by FFS, which is also computationally efficient. Chi-squared is a flexible tool for analyzing relationships in data because it can be used to compare two or more categorical variables.

Model

In this work we have used Random Forest, decision trees, neural networks, SVM, linear and logistic regression to train our dataset. All the models work especially well with high-dimensional data.

III. EVALUATION

For Evaluation, the grid search method has been used for this model's Linear regression, Logistic regression, SVM, Decision trees, Random Forest, Multi-layer Perceptron, Gradient Boosting. Grid search method used to find the best hyperparameters for each model which gives optimal result for the dataset.

Classification

The models used for classification are:

Logistic Regression

A grid search over the values of five hyperparameters for a logistic regression. After training the model with correlation & forward feature selection the best hyperparameters are C: 1, max_iter: 10000, penalty: l2, random_state: 42, solver: liblinear.

The logistic regression model performs slightly better with the correlation-based features than the forward based features. The model has a balanced accuracy of 0.584 (correlation) and 0.469 (forward) with an overall accuracy of 0.79. The model also has a precision of 0.83, a recall of 0.92 and a F1-score of 0.87 (correlation) and a precision of 0.83, a recall of 0.25 and a F1-score of 0.32 (forward).

SVM

A grid search over the values of four hyperparameters for SVM. After Training the model, with correlation & forward feature selection the best hyperparameters are Correlation C: 100, gamma: 0.01, kernel: rbf, random_state: 42.

The SVM model performs similarly to the logistic regression model, with a balanced accuracy of 0.627 (correlation) and 0.5 (forward) and an overall accuracy of 0.75. The model also has a precision of 0.85, a recall of 0.84 and a F1-score of 0.85 (correlation) and a precision of 0.85, a recall of 0.42 and a F1-score of 0.40 (forward).

Decision Tree Classifier

A grid search over the values of five hyperparameters for an Decision Tree. After Training the model, with correlation & forward feature selection the best hyperparameters. criterion: gini, max_depth: 10, min_samples_leaf: 1, min_samples_split: 2, random_state: 42**.**

Both the correlation and forward models perform well. They have a balanced accuracy of 0.52 and with a similar accuracy.The correlation model has a slightly higher precision (0.81) for the positive class than the forward model (0.22). The forward model has a slightly higher recall (0.33) for the positive class than the correlation model (0.71). The overall performance of the correlation and forward models is similar.

Random Forest Classifier

Hyperparameters for a random forest classifier, After training, with correlation & forward feature selection the best hyperparameters for correlation max_depth: 10, min_samples_leaf: 1, min_samples_split: 10, n_estimators: 200, random_state: 42.

The balanced accuracy of the model is 0.59. The classification report shows that the model has a precision of 0.84, a recall of 0.94, and an F1-score of 0.88. The overall accuracy is 0.80. Due to imbalanced dataset, we used oversampling methods like, SMOTE & Random oversampling and got balanced accuracy of 83% using random oversampler.


```
In [263]: # Perform 10-fold cross-validation on the dataset using the Random forest
k = KFold(n_splits=10)
scores = cross_val_score(rf_classifier, X_train, y_corr, cv=k)
# Printing the accuracy of each fold
print("Accuracy for each fold:")
for i in range(len(scores)):
    print("Fold " + str(i) + ": " + str(scores[i]))

# Printing the average accuracy
average_accuracy = sum(scores) / len(scores)
print("Average accuracy: " + str(average_accuracy))

Accuracy for each fold:
Fold 0: 0.8064516129032258
Fold 1: 0.9032258064516129
Fold 2: 0.8064516129032258
Fold 3: 0.7741935483870968
Fold 4: 0.7
Fold 5: 0.8
Fold 6: 0.8333333333333334
Fold 7: 0.6666666666666666
Fold 8: 0.7
Fold 9: 0.8666666666666667
Average accuracy: 0.7856989247311829
```

Fig-1(The classification accuracy of Random forest while predicting PCR).

MLP Classifier

Hyperparameters for a neural network classifier, After training, with correlation & forward feature selection the best hyperparameters for forward activation: relu, hidden_layer_sizes: [32, 16, 8], solver: sgd.

The model has a balanced accuracy of 0.47, which means that the model correctly classifies 47.4% of the data. The classification report shows that the model has a precision of 0.90, a recall of 0.78, and an F1-score of 0.83 for class 0, and a precision of 0.08, a recall of 0.17, and an F1-score of 0.11 for class 1. The overall accuracy is 0.72.

GradientBoostingClassifier

Hyperparameters for a gradient boosting classifier,

After Training, with correlation & forward feature selection the best hyperparameters learning_rate: 0.1, max_depth: 5, min_samples_leaf: 1, min_samples_split: 10, n_estimators: 300, random_state: 42.

The model has a balanced accuracy of 0.45. The classification report shows that the model has a precision of 0.79, a recall of 0.92, and an F1-score of 0. The overall accuracy is 0.74.

Regression

The models used for Regression are:

Linear Regression

Hyperparameters for linear regression, after training, with correlation & forward feature selection the best hyperparameters copy_X: True, fit_intercept: True, n_jobs: 1.

The Forward model had an MAE of 22.10, which is also lower than the correlation model's MAE of 23.45.

SVM

Hyperparameters for SVM regression, After training, with correlation & forward feature selection the best hyperparameters C: 10, epsilon: 1.0, gamma: 0.01, kernel: rbf.

The SVM regression model trained on the forward achieved a lower MAE value of 21.96 compared to the SVM regression model trained on the correlation 23.80.

Decision Tree

Hyperparameters for Decision Tree regression, After training, with correlation& forward feature selection the best hyperparameters criterion: squared_error, max_depth: 2**,** min_samples_leaf: 1, min_samples_split: 2, random_state: 42

splitter: random.

the decision tree regressor trained on the correlation dataset had an mae of 23.45 & forward has mae of 22.10. Comparing the two MAE values, we can see that the decision tree regressor trained on the forward dataset had a lower MAE value than the decision tree regressor trained on the correlation dataset.

MLP Regressor

Hyperparameters for MLP regression, After training, with correlation & forward feature selection the best hyperparameters activation: tanh, alpha: 0.01, hidden_layer_sizes: (40,), learning_rate: constant, random_state: 0, solver: sgd.

The MLP regression model trained on the correlation dataset had an mae of 23.87 & forward dataset has mae of 24.18. Comparing the two MAE values, we can see that the MLP regression model trained on the correlation dataset had a lower MAE value than the MLP regression model trained on the forward dataset.

GradientBoostingRegressor

Hyperparameters for Gradient Boosting regression, After training, with correlation & forward feature selection the best hyperparameters learning_rate: 0.05, loss: squared_error, max_depth: 10, min_samples_split: 10, n_estimators: 100, random_state: 42, subsample: 0.75.

The gradient boosting regressor model trained on the correlation dataset had an mae of 22.35 & forward dataset had an mae of 22.60. Comparing the two MAE values, we can see that the gradient boosting regressor model trained on the correlation dataset had a lower MAE value than the gradient boosting regressor model trained on the forward dataset.

Random Forest Regressor

Hyperparameters for Random Forest regression, After training, with correlation & forward feature selection the best hyper-parameters, n_estimators:100, max_depth:9, random_state:10.

The random forest Regressor model trained and an mean absolute error of 15.55.

```
]: # Evaluate the performance of the model
   mae = mean_absolute_error(y_test, y_pred)
   print(f'Mean Absolute Error: {mae}')

   Mean Absolute Error: 15.558959021247993

]: k_fold = KFold(n_splits=10, shuffle=True, random_state=10)
   mae_scores = cross_val_score(rf_regressor, new_X, y, scoring='neg_mean_absolute_error', cv=k_fold)

]: for i, mae in enumerate(mae_scores, 1):
       print(f'Fold {i}: Mean Absolute Error: {abs(mae)}')

   Fold 1: Mean Absolute Error: 15.95452099047807
   Fold 2: Mean Absolute Error: 22.744015008953035
   Fold 3: Mean Absolute Error: 22.564765773286915
   Fold 4: Mean Absolute Error: 21.658076177302416
   Fold 5: Mean Absolute Error: 17.86325355765368
   Fold 6: Mean Absolute Error: 17.77127889652033
   Fold 7: Mean Absolute Error: 18.172086566143612
   Fold 8: Mean Absolute Error: 22.148916568258475
   Fold 9: Mean Absolute Error: 19.12241858102302
   Fold 10: Mean Absolute Error: 22.582866497002716

]: print(f'Average Mean Absolute Error across all folds: {abs(mae_scores.mean())}')

   Average Mean Absolute Error across all folds: 20.058219861662227
```

Fig-2(Mean absolute error of random forest regression while predicting RFS)

IV. Discussion

Classification

The best performing model for classification task was Random Forest Classifier. This model was highly accurate and also managed to balance precision and recall effectively. The Random Forest model outperforms the other models in terms of balanced accuracy and overall accuracy, making it the most reliable choice for this specific task.

Advantages of Random Forest Classifier

Random forests are known for their high accuracy in predicting, handling large datasets and complexities. Its relatively robust to outliers and noise [4]. They can also handle data imbalance and missing data and do not require input variables to be scaled which made it easy to work with in general.

Disadvantages of Random Forest Classifier

They are known to get really complex really quickly. Due to an ensemble nature of model (creating multiple trees), random forests can consume a substantial amount of memory which can get very challenging for very large datasets. For problems where the relationship between variables is linear, simpler models like linear regression may perform better than Random Forests.

Regression

We employed Gradient boosting, MLP, Decision Trees, SVR (Support Vector Regression), Linear Regression and out of all these Gradient Boosting Regressor with Forward feature selection was the best performing model. It had the lowest MSE and low MAE showing it was most effective in predicting with minimal error.

Advantages of Gradient Boosting Regression

Gradient boosting belongs to the ensemble methods category, wherein multiple weak models are generated and then combined to enhance overall performance [5]. It gives us high performance (often in prediction tasks), handles both continuous and categorical features and also gives us an insight over the most influential features in predictions. Within gradient boosting machines, the learning process involves sequentially fitting new models to improve the accuracy of estimating the response variable [6].

Disadvantages of Gradient Boosting Regression

Can be computationally complex, usually being slow during training. In cases of noisy data, we risk overfitting by using this. This model is also very sensitive to hyperparameter settings. So, careful tuning is advised.

Other Models

SVM with a balanced accuracy has shown promise with appropriate kernel and regularization parameters in handling class imbalance and Linear Regression's performance could possibly be improved using stacking or bootstrapping. Other models like MLP, decision trees have performed moderately against the given dataset.

V. CONCLUSION

After training each model for both classification and regression with different datasets and hyperparameters, the best model we obtained using the grid search method for classification was a random forest classifier for a correlated dataset with a balanced accuracy of 60% and a model accuracy of 87%. Due to the imbalanced data in the dataset, we further attempted to improve the balanced accuracy by using oversampling methods such as SMOTE and Random Oversampling. The highest balanced accuracy of 83% was achieved using the random Oversampling method and an overall accuracy of 78%.

For regression, the best model we obtained using grid search was a random forest regressor using a forward feature-selected dataset. The Mean Absolute Error achieved for this model was 15.5, and the overall K-fold error was 20.

In the future, we can train the classification model with a balanced dataset and a more complex neural network model using hyperparameters such as early stopping, learning rate decay, and ReduceLROnPlateau to obtain better results and prevent overfitting.

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